



Graduation Project · Final Presentation · Team 0261

Lingo

**Interactive AI System for Handwriting
Recognition & Language Enhancement**

lingo-handwriting-recognition.netlify.app

A 10-person team · Web · PWA · Android





ten people, one product

The team

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with thanks

Guided by experts.

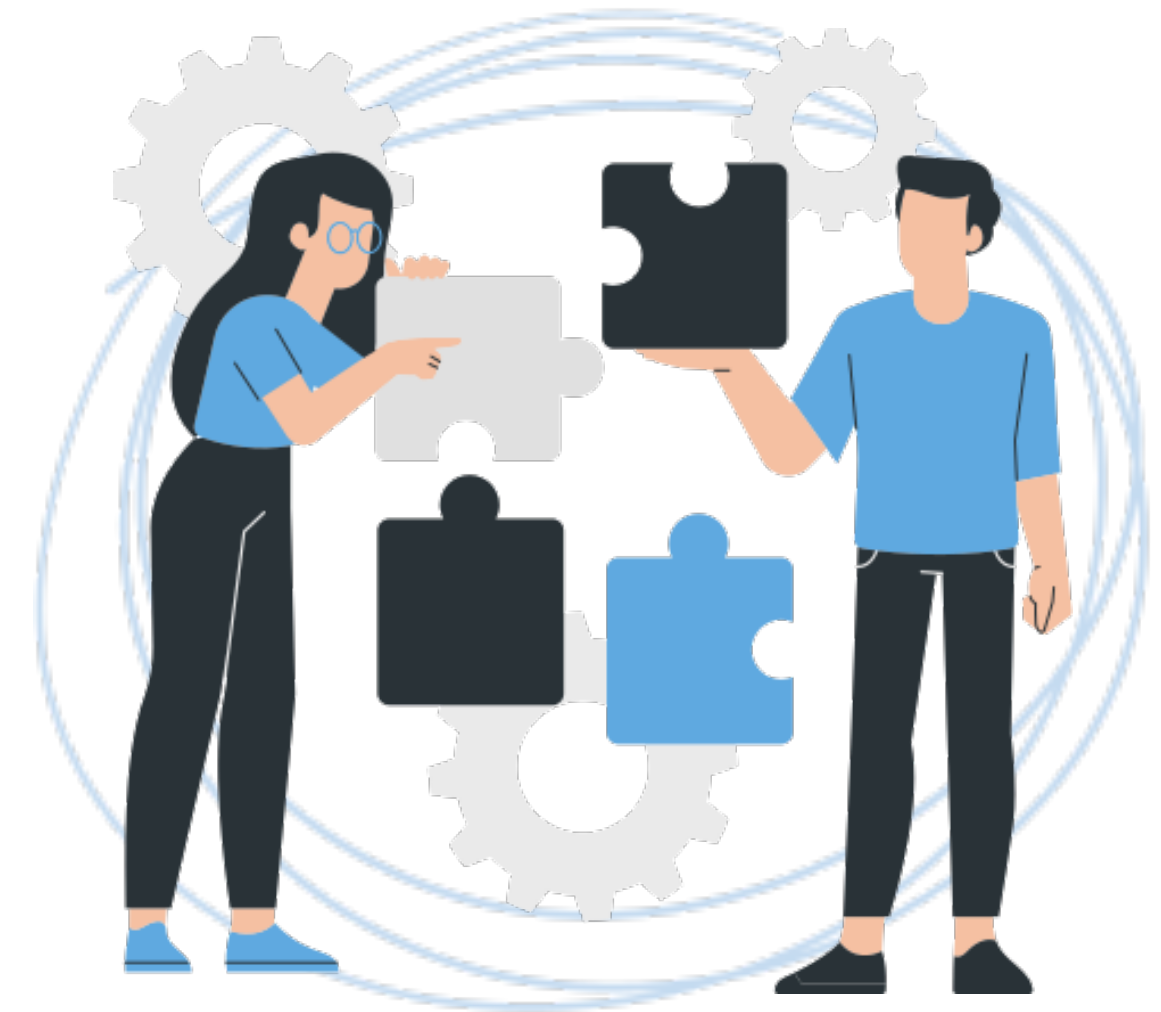
This project owes a great deal to the support, insight, and patience of our two mentors throughout the year.

Dr. Mahmoud Bassiouni

Project Mentor

Eng. Nader Emad

Technical Mentor



Chapter 01

Introduction

01



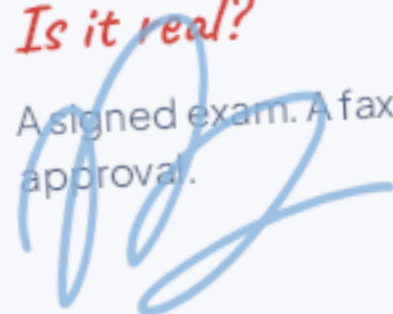


the reality today

Every day, real decisions are made on
handwritten paper that **nobody can**
actually verify.

Is it real?

A signed exam. A faxed
approval.



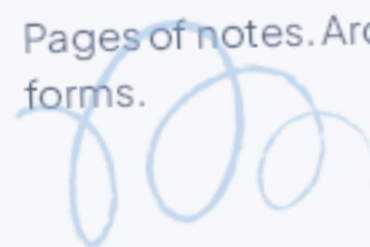
Is it correct?

A student's English. A child's
homework.

teh quike brown

What does it say?

Pages of notes. Archived
forms.



Chapter 02

Problem Definition



the gap we found

Handwriting is everywhere, yet unreadable at scale.

Hard to read

Stacks of handwritten pages cannot be searched, edited, or shared digitally without slow manual typing.

Hard to check

No quick way to flag spelling and grammar mistakes, or to know which language a page is even written in.

Easy to forge

Signatures on exams, approvals, and contracts are trusted by eye alone, with no reliable check for fakes.

No single tool today will **read** a handwritten page, **check** its language, and **verify** its authenticity. That gap is the problem we set out to solve.



Objectives

What we set out to build

01 Digitize handwriting accurately

Convert any handwritten page into clean, editable, searchable text.

02 Detect & check language

Identify the language automatically and flag spelling and grammar issues.

03 Verify signatures

Build our own deep-learning model to tell genuine signatures from forgeries.

04 Interact with documents

Let users ask questions and get answers grounded in their own files.

Chapter 03

Target Audience

03





"are these signatures real?"

THE PROFESSOR

Hundreds of signed papers. Zero time to check them.

their world

Signed exam papers, faxed approvals, handwritten submissions, and no fast way to tell what's genuine or read it back later.

what Lingo gives them

Confirm a signature is authentic in seconds, and turn every handwritten record into searchable, exportable text.

THE LANGUAGE LEARNER

"Did I write this correctly?"

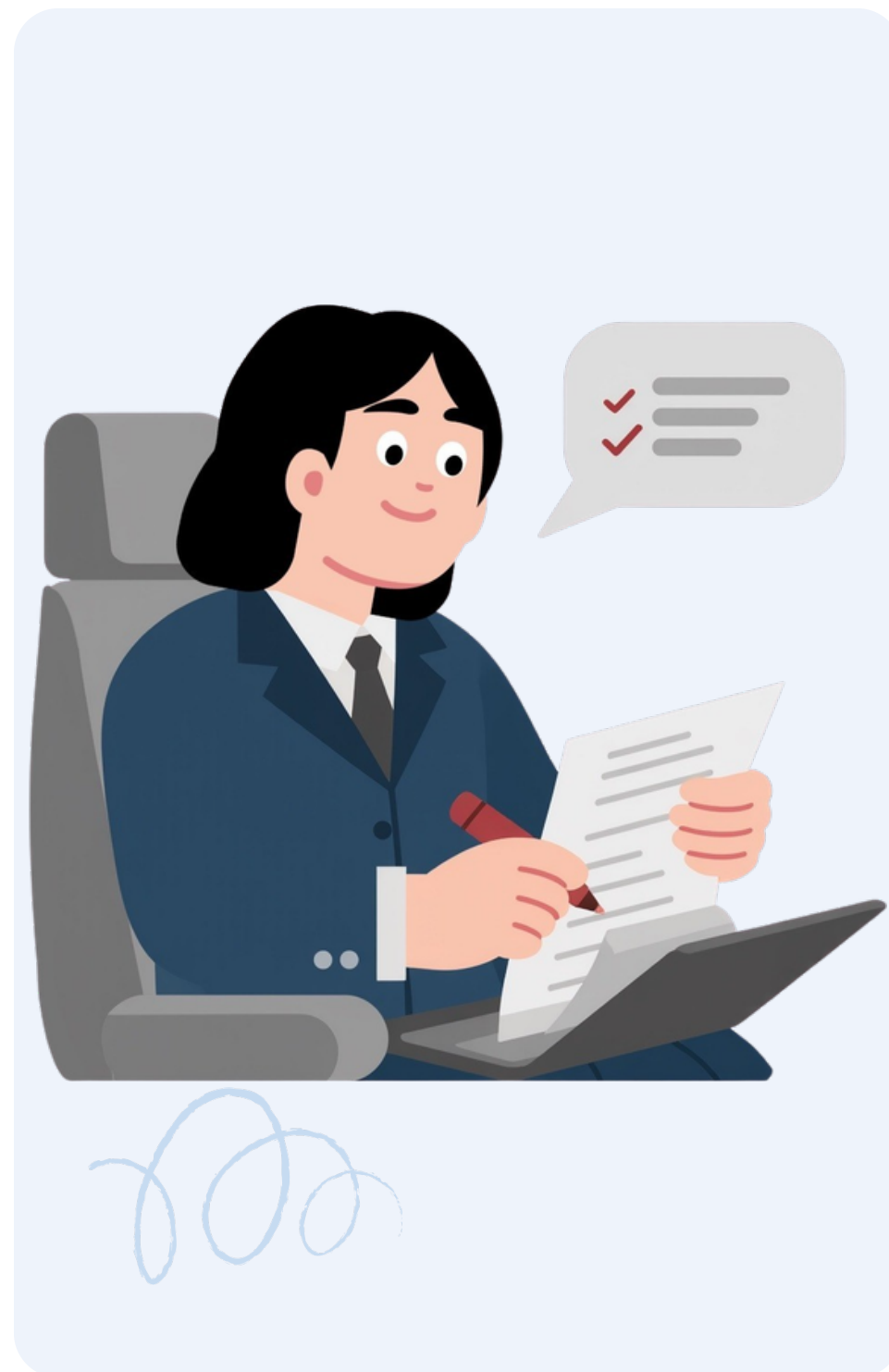
their world

A student practising English or Arabic. A parent checking a child's homework. Handwriting that mixes both scripts on one page.

what Lingo gives them

Snap a photo and get the language detected automatically plus feedback on spelling and grammar, English, Arabic, or mixed.





THE ADMINISTRATOR · BANK · HR · REGISTRY

Thousands of forms. Fraud on any page.

their world

Archives of signed contracts and forms. Real fraud exposure, and nobody with time to read all of it.

what Lingo gives them

Flag forged signatures, digitize the whole archive, and ask plain-language questions across thousands of documents.

Chapter 04

Proposed Solution





one system that lets you...

Trust it. Prove a signature is genuine.

Understand it. Check the language, spelling & grammar.

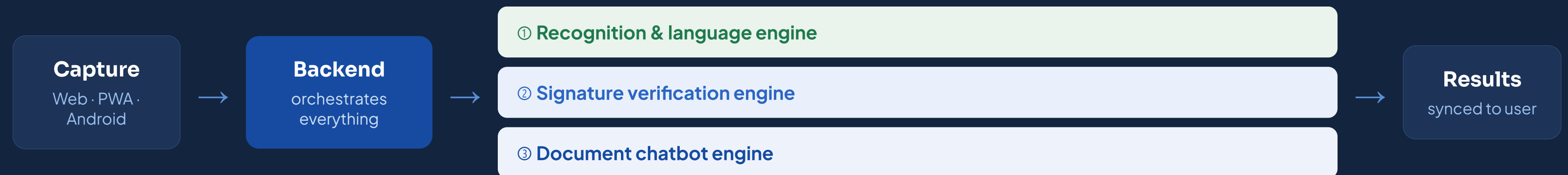
Talk to it. Ask questions across your documents.



Turning the writing into editable text is the **front door**. Everything that makes Lingo valuable happens **after** the page is digital.

System Flow

One capture, the whole system



Three **different** AI engines, each solving a different problem, behind one simple capture-and-go workflow. We will look at all three in detail.

Chapter 05 · the core of our work

The AI Models

05





meet the three models

Three engines, three jobs.

model 01 · the core

Recognition & Language

Reads the page, detects the language, and checks spelling & grammar.

model 02 · trained in-house

Signature Verification

Tells a genuine signature from a forgery with a deep-learning model we trained.

model 03 · the assistant

Document Chatbot

Answers questions grounded in the user's own uploaded documents.

A six-stage AI pipeline

1 Input Acquisition

User uploads a handwritten image or document.

Data Acquisition

2 Handwriting Recognition

Detects handwritten characters and words.

Computer Vision + Deep Learning

3 Language Detection

Determines the text language for correct processing.

Natural Language Processing

4 Text Enhancement

Corrects spelling and grammar.

NLP / Language Processing

5 Notes Generation

Generates human-readable notes from the text.

NLP / Explainability

6 Accuracy Evaluation

Measures similarity between original and corrected text.

Text Similarity Metrics

The process

Layer (type)	Output Shape	Param
InputLayer	(None, 200, 50, 1)	
Conv_1 (Conv2D)	(None, 200, 50, 64)	64
Conv_2 (Conv2D)	(None, 200, 50, 32)	18,464
MaxPooling2D	(None, 100, 25, 32)	
Conv_3 (Conv2D)	(None, 100, 25, 64)	18,496
Conv_4 (Conv2D)	(None, 100, 25, 32)	18,464
MaxPooling2D_1	(None, 50, 12, 32)	
Flatten_layer (Reshape)	(None, 50, 384)	
Dense_1 (Dense)	(None, 50, 64)	24,640
Dense_2 (Dense)	(None, 50, 64)	4,160
Dropout	(None, 50, 64)	
BiDirectional_lstm_1 (BiDirectionalLSTM)	(None, 50, 256)	197,632
BiDirectional_lstm_2 (BiDirectionalLSTM)	(None, 50, 128)	164,352

```
img_2_path = "/kaggle/input/handwriting-recognitionocr/test_images/img_2.png"
img_2 = plt.imread(img_2_path)
plt.imshow(img_2)
print(single_sample_prediction(pred_model, img_2_path))
```

/1 _____ 0s 19ms/step
 'JULES']

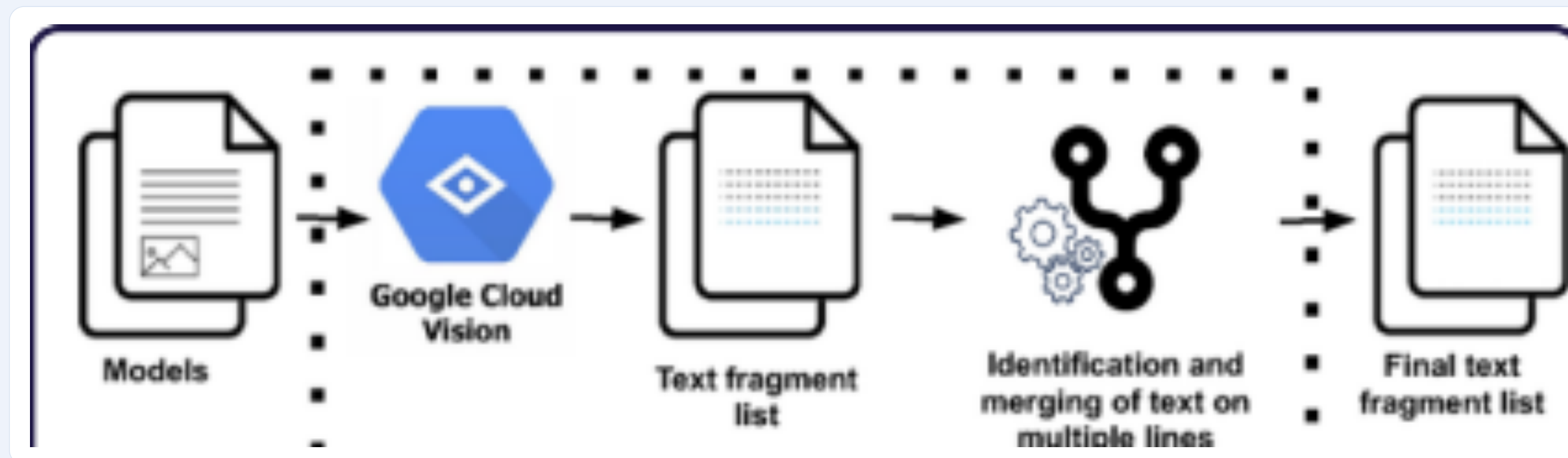


It worked, the model correctly read the handwritten word "JULES."

CRNN architecture, convolutional layers feeding bidirectional LSTMs.

Production-grade capture with Google Cloud Vision

- ✓✓✓✓ Trained on **large-scale, diverse datasets**, no need for our own massive dataset.
- ✓✓✓✓ Higher, **more stable accuracy** across messy real-world handwriting.
- ✓✓✓✓ Deep-learning text detection, then merges text across multiple lines.



95%

accuracy on handwritten text,
under proper conditions.

engineering maturity

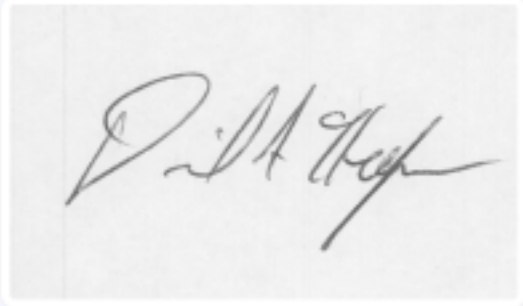
We built our own, learned its limits, and
chose the right tool for production.

Is this signature real?

Signature Verification

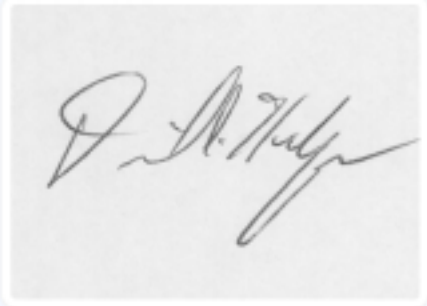
Reference signature

A known-genuine signature for comparison.



Signature to verify

The signature whose legitimacy you want to check.



Verify signature

Reset

✔

Likely genuine

Similarity score: 99.99%

why deep learning?

Verification is an image-similarity task. Deep learning learns high-level features **directly from images**, instead of relying on hand-crafted rules.

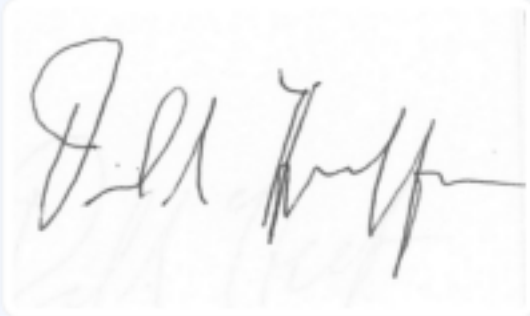
custom dataset

Built by combining public signature datasets, primarily the **CEDAR** dataset, with both **genuine** and **forged** samples for training and testing.

Signature Verification

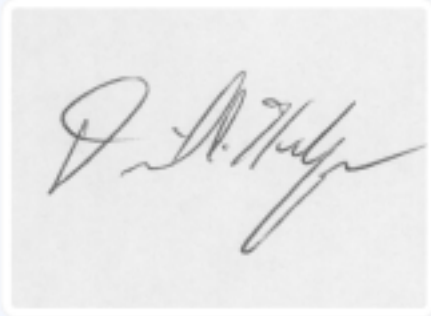
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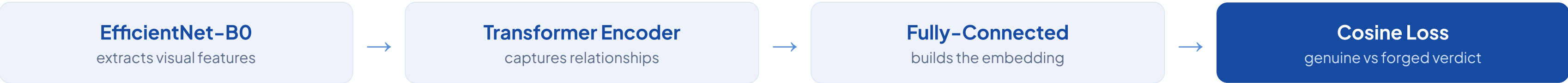
✘

Likely forged

Similarity score: 26.18%

MODEL 02 · SIAMESE TRANSFORMER NETWORK

How the model works



training configuration

Image size 224 × 224	Batch size 32	Epochs 40
Optimizer AdamW	Learning rate 0.0001	Data split 80 / 20

Training accuracy

96.33%

Test accuracy

93.68%

An assistant that has read your documents

what it can do

- ✓✓✓✓ Reads images and files, and extracts the text automatically.
- ✓✓✓✓ Answers questions and keeps conversation context.
- ✓✓✓✓ Analyzes documents and gives summaries and recommendations.

supported files

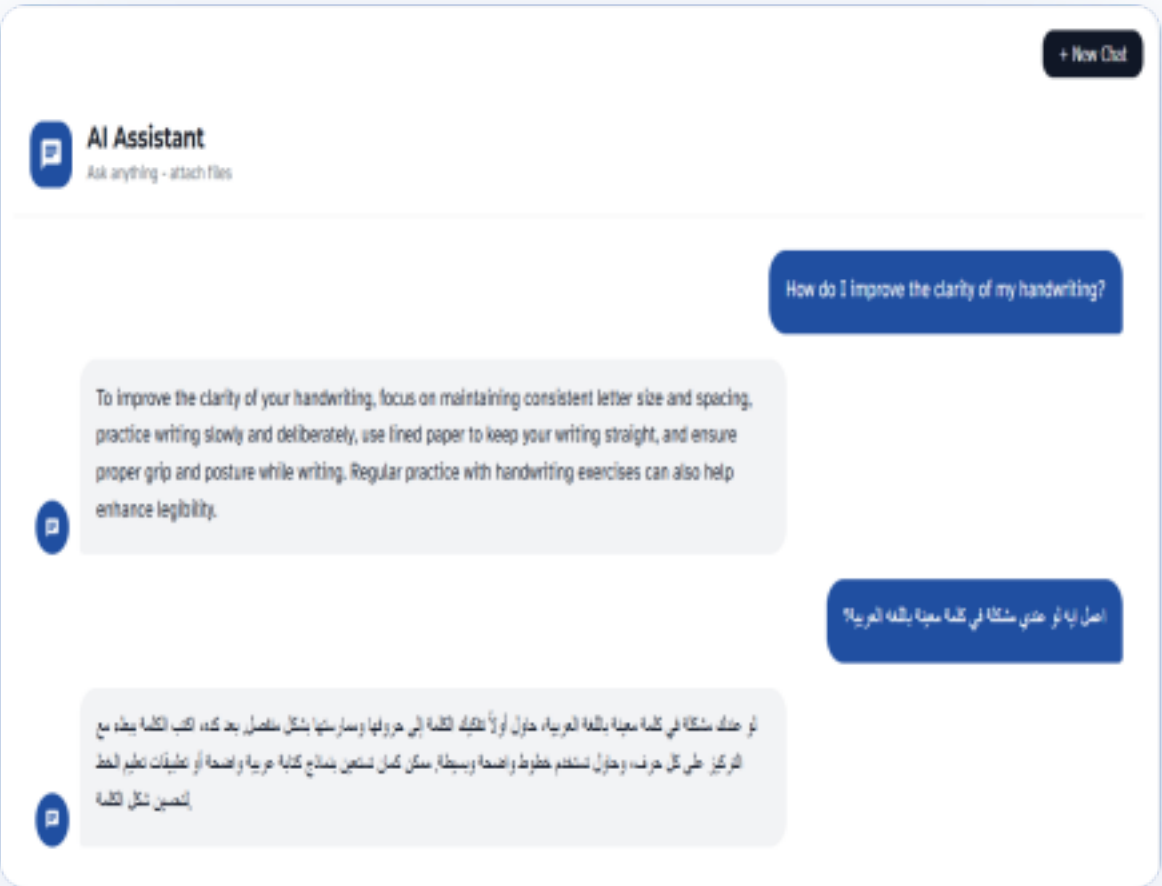
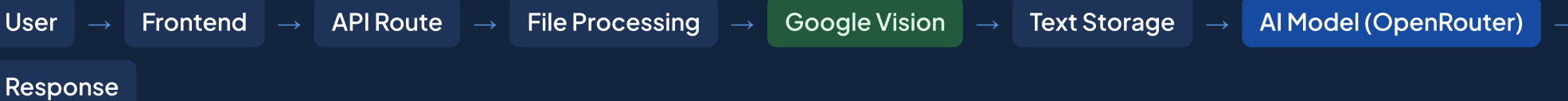
JPG · PNG · JPEG

PDF

PPTX

♦ **All languages supported** — Arabic, English, and more.

system architecture



MODEL 03 · WHAT MAKES IT SMART

An engineered AI Chatbot assistant

Task Detection

Classifies each question: recognition, accuracy, forgery, improvement, or general.

Emotion Detection

Reads positive, neutral, or frustrated to adapt the response style.

Conversation Memory

Retains the last **10 messages** to understand context.

Context Retrieval

Splits large files and picks the most relevant chunks, fewer tokens, faster, cheaper.

Prompt Engineering

Builds the prompt from task, emotion, text, context and memory.

Session & Security

Per-user key and session, with protection against prompt-injection.

Chapter 06

Frontend

06



Frontend · how we built the interface

Designed in Figma, built in Next.js

1 · Design in Figma

wireframe & prototype every screen

2 · Build components

reusable React + TypeScript

3 · Make it responsive

five device sizes

4 · Ship as a PWA

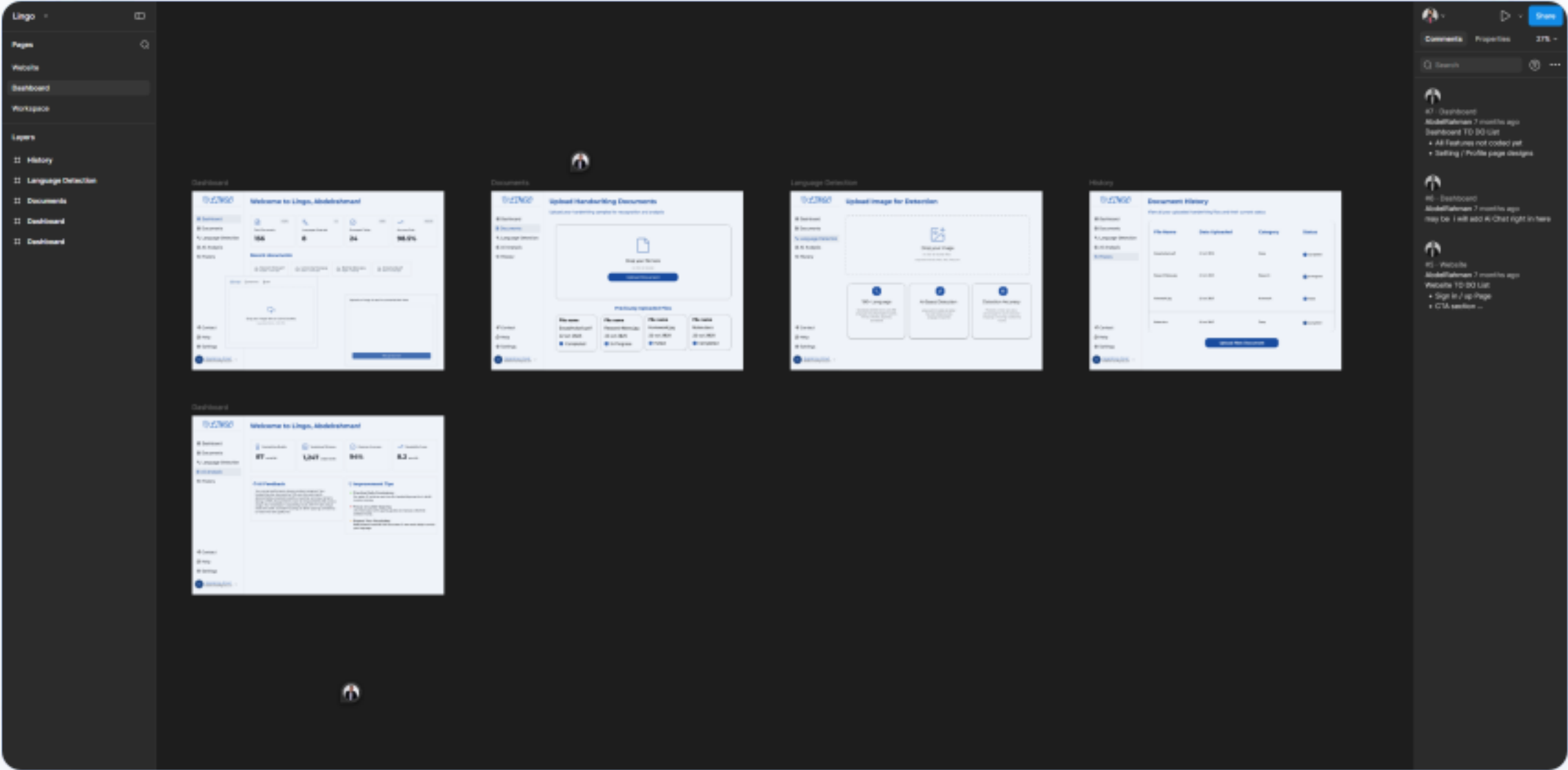
installable, works offline

Figma

Next.js

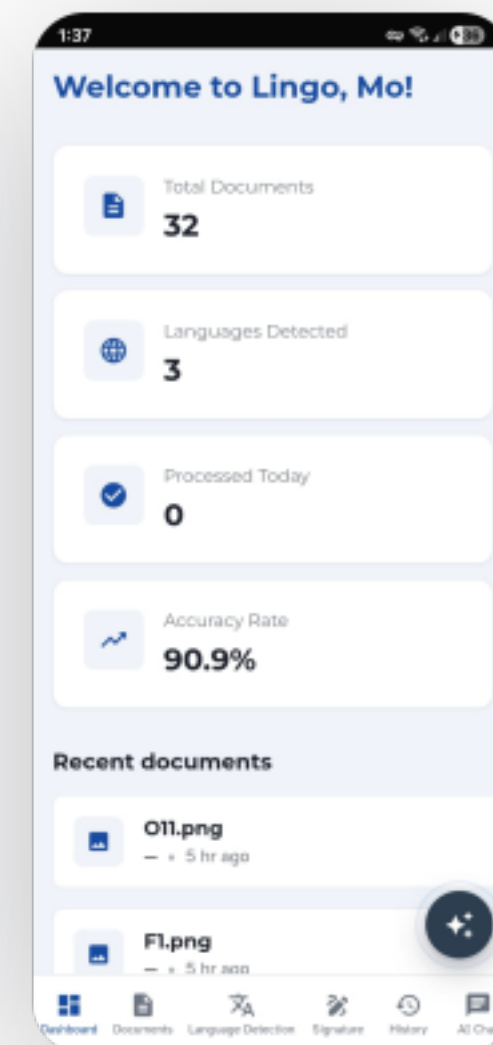
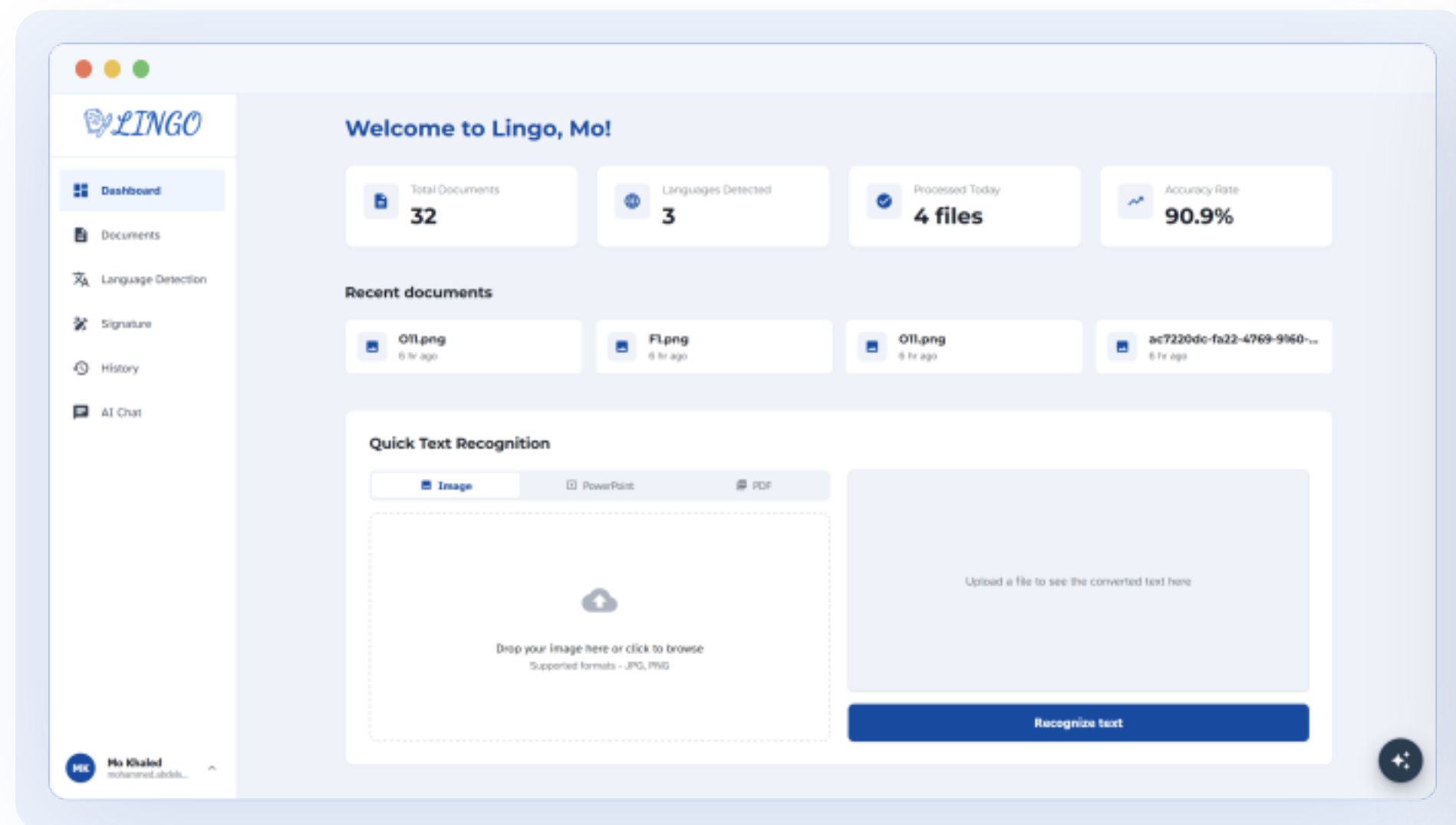
React

TypeScript



Frontend · the result

A real, shipped dashboard



Responsive

Lays out cleanly across five device sizes, phone to desktop.

Installable PWA

Add to home screen and launch like a native app.

Offline-tolerant

Keeps working through poor or dropped connections.

one codebase, every screen

Wherever the paper is, **Lingo** is.

Web

Responsive dashboard in any browser, on any device.

PWA

Installable app that works offline, no store needed.

Android

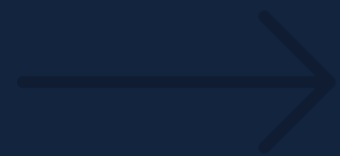
Signed APK, the same power in your pocket.

One codebase builds all three. Build it once, it runs everywhere.

Chapter 07

Backend

07



Backend · the orchestrator

A hybrid architecture

Node.js on Netlify

API routes handle standard application logic and fast request/response work.

Python on Firebase

Cloud Functions run the heavy machine-learning, including the PyTorch signature model.

The frontend **never talks to the AI directly**. The backend sits in between, which protects our API keys and lets heavy tasks run asynchronously without freezing the user's device.

Backend · orchestration flow

Five steps, end to end

- | | | |
|----------|--------------------------------|---|
| 1 | Ingestion & Trigger | Frontend uploads a file and triggers the backend. |
| 2 | AI Recognition Chain | Backend runs recognition through the Vision and NLP services. |
| 3 | Heavy ML Routing | Routes the heavy work to the PyTorch signature model. |
| 4 | Context-Aware Chatbot | Powers the chatbot through the OpenRouter language model. |
| 5 | Real-Time Sync | Syncs results back to the database and the user. |

Chapter 08

Results & Roadmap



everything under the hood

Technology stack

Frontend

Next.js
React
TypeScript
Figma

Backend

Node.js · Netlify
Python · Firebase
Cloud Functions
JSON storage

AI / ML

PyTorch
EfficientNet-B0
Google Cloud Vision
OpenRouter LLM

Processing

PDF Parse
JSZip (PPTX)
Session management
Firebase Auth

not a demo, a deployed product

Live in production today.

One codebase, three platforms

Web, installable PWA, and a signed Android APK ship from the same source.

Your documents stay yours

Per-user security rules mean you only ever see your own files.

Costs nothing when idle

The AI engines scale to zero, then spin up instantly on demand.

Works on any screen, even offline

Responsive across five device sizes with offline-tolerant caching.

where we take it next

What's next

01

Premium subscription tier

Unlock advanced features for paid users, including higher usage limits and priority processing.

02

More AI choice

Add more providers behind the assistant so it picks the best model per task.

03

On the Play Store

Publish the signed Android app for one-tap install by anyone.



enough slides,

See it on real paper.



Live demo & Android app
lingo-handwriting-recognition.netlify.app





Trust it.
Understand it.
Talk to it.

Thank you. We'd love your questions.

lingo-handwriting-recognition.netlify.app

Interactive AI System for Handwriting Recognition & Language Enhancement

